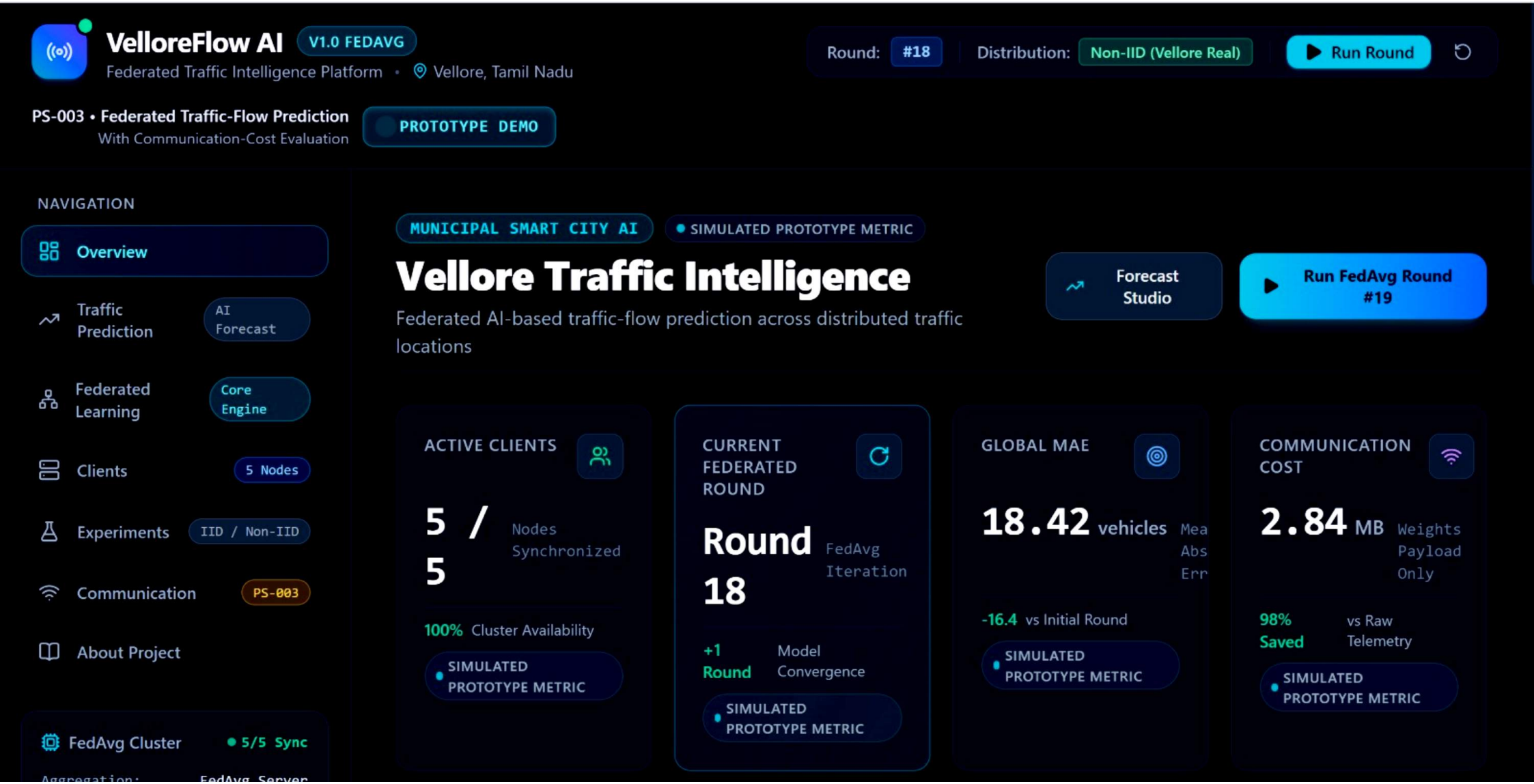

VelloreFlow AI
V1.0 FEDAVG
Federated Traffic Intelligence Platform • Vellore, Tamil Nadu
Round: #18
Distribution: Non-IID (Vellore Real)
Run Round
PS-003 • Federated Traffic-Flow Prediction
With Communication-Cost Evaluation
PROTOTYPE DEMO
NAVIGATION
Overview
Traffic Prediction
AI Forecast
Federated Learning
Core Engine
Clients
5 Nodes
Experiments
IID / Non-IID
Communication
PS-003
About Project
FedAvg Cluster
5/5 Sync
MUNICIPAL SMART CITY AI
SIMULATED PROTOTYPE METRIC
Vellore Traffic Intelligence
Federated AI-based traffic-flow prediction across distributed traffic locations
Forecast Studio
Run FedAvg Round #19
ACTIVE CLIENTS
5 / 5
Nodes Synchronized
100% Cluster Availability
SIMULATED PROTOTYPE METRIC
CURRENT FEDERATED ROUND
Round 18
FedAvg Iteration
+1 Round
Model Convergence
SIMULATED PROTOTYPE METRIC
GLOBAL MAE
18.42 vehicles
Mea Abs Err
-16.4 vs Initial Round
SIMULATED PROTOTYPE METRIC
COMMUNICATION COST
2.84 MB
Weights Payload Only
98% Saved
vs Raw Telemetry
SIMULATED PROTOTYPE METRIC

CamScanner

VelloreFlow AI

V1.0 FEDAVG

Federated Traffic Intelligence Platform • Vellore, Tamil Nadu

Round: #18 | Distribution: Non-IID (Vellore Real) | Run Round

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PROTOTYPE DEMO

Payload / Node: 158 KB weights

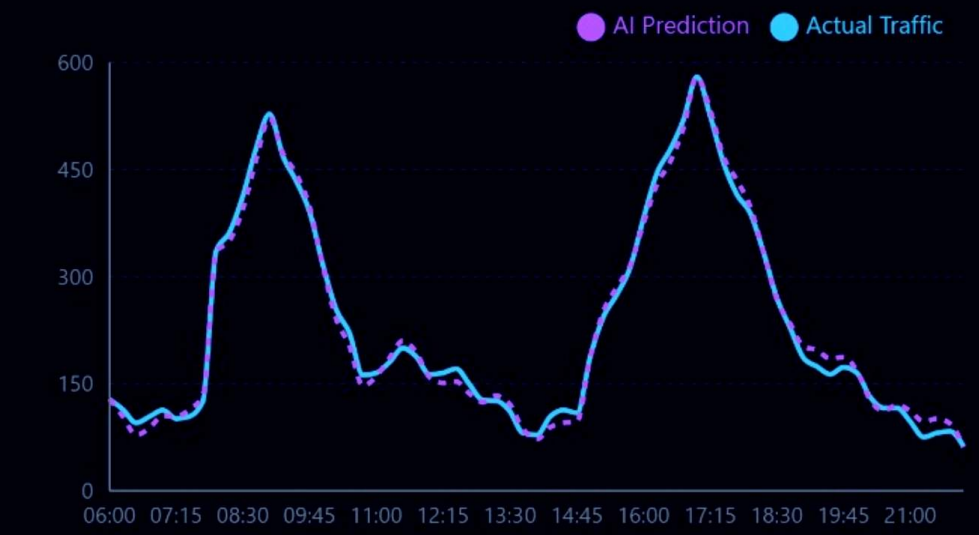
Distribution: Vellore Non-IID

Traffic Prediction vs Actual Traffic

FedAvg Global Model

Katpadi Junction

Timestamped vehicle counts across morning and evening peak intervals in Vellore



Morning Peak: 08:30 - 10:15 Evening Peak: 17:30 - 20:00

SYNTHESIZED FROM VELLORE TRAFFIC SERIES

Federated Learning Status

Round 18

Real-time synchronization cycle between Vellore traffic edge clients and FedAvg server

Round 18
Federated epoch initialized

5 clients participated
Katpadi, Green Circle, Old Bus Stand, Bagayam, Collectorate

Local training completed
On-device SGD/Adam optimization on edge

Model updates received
5 × 158 KB model weights uploaded via gRPC

FedAvg aggregation completed
Normalized weighted parameter averaging

Global model updated
New model broadcast to edge nodes for inference

CamScanner

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PS-003 • Federated Traffic-Flow Prediction
With Communication-Cost Evaluation

PROTOTYPE DEMO

Distributed Edge Client Status (5 Nodes)

Each simulated client processes localized traffic observations without sharing raw data

Detailed Client Registry →

Client 1	Client 2	Client 3	Client 4	Client 5
Online	Online	Online	Online	Online
Vellore Traffic...	Vellore Traffic...	Vellore Traffic...	Vellore Traffic...	Vellore Traffic...
Local Samples: 8,420	Local Samples: 7,850	Local Samples: 9,130	Local Samples: 7,310	Local Samples: 8,060
Local MAE: 19.2	Local MAE: 17.9	Local MAE: 18.7	Local MAE: 20.1	Local MAE: 18.3
Update Size: 158 KB	Update Size: 158 KB	Update Size: 158 KB	Update Size: 158 KB	Update Size: 158 KB
Distribution: Non-IID	Distribution: Non-IID	Distribution: Non-IID	Distribution: Non-IID	Distribution: Non-IID
Round 18 • 2m ago	Round 18 • 2m ago	Round 18 • 2m ago	Round 18 • 2m ago	Round 18 • 2m ago

Hackathon PS-003

Municipality of Vellore traffic observations treated as 5 distributed edge clients with communication cost benchmarking.

Demo Mode Notice: Prototype demonstration using simulated traffic predictions and federated-learning metrics. Final experimental results will be generated from the actual Vellore dataset and trained model.

Zero Live APIs • Static Simulation

Vellore Municipal Edge Nodes

CamScanner